detection performance of Short-PHD is improved. However, note that a PHD distribution does not strictly follow a normal distribution. These statistical results should only be considered an approximation.

4 Experiments

In this section, we evaluate our Short-PHD method using both public and generated datasets. We benchmark our method against various zero-shot and supervised detection approaches. Additionally, we conduct multiple sensitivity analyses for Short-PHD and evaluate its robustness against several detection attacks.

4.1 Experimental Setup

Public Dataset Setting. We first evaluate the Short-PHD method on the original dataset used in the PHD method (Tulchinskii et al., 2024), referred to as GPTID. The dataset consists of two domains of human-written text: Wiki40b (Guo et al., 2020) from Wikipedia and WritingPrompts from Reddit's stories subsector (Fan et al., 2018). The paper claims another StackExchange subset, but it is not publicly available. Text generation is performed using GPT-2 XL (Radford et al., 2019), OPT-13B (Zhang et al., 2022), and GPT-3.5. Detailed statistics are summarized in Table A1.

Generated Dataset Setting. We then construct a more comprehensive dataset in two additional domains using more diverse and advanced LLMs to further evaluate the performance of our method. Specifically, we select Wikihow (Koupaee & Wang, 2018) for articles and quizzes and Eli5 (Fan et al., 2019) for question answering. We employ OPT-13B (Zhang et al., 2022), OPT-iml-1.3B (Iyer et al., 2022), GPT-NeoX (Black et al., 2022), GPT-3.5-turbo, and GPT-4o for text generation. For each domain, we randomly sample 500 human-written texts and prompt each LLM to generate text. Detailed generation settings are provided in Appendix B.1.

Baselines. For zero-shot methods, we mainly compare our Short-PHD method with other state-of-the-art approaches, including PHD (Tulchinskii et al., 2024), DetectGPT (Mitchell et al., 2023), and DetectLLM (Su et al., 2023). Additionally, we evaluate Short-PHD against methods using heuristic scores, including Perplexity (mean log probabilities) and LogRank (average log of ranks). For supervised methods, we incorporate a fine-tuned RoBERTa model trained on GPT-2 outputs from OpenAI (Solaiman et al., 2019).

Implementation Details. We truncate human-written and LLM-generated texts to 50, 100, and 150 tokens to evaluate detection performance for short texts. For the Short-PHD method, we perform OCI 12 times using different off-topic content and utilize LLaMA-3-8B (Dubey et al., 2024) to compute the embedding of each insertion (i.e., the 5th line in Algorithm 1). For a fair comparison, we also leverage LLaMA-3-8b as the scoring model for all zero-shot baselines. The RoBERTa model is directly taken from OpenAI's public version without further modification. More implementation details are provided in Appendix B.1.

4.2 Main Results

Table 3: Results on the GPTID dataset. The table reports the AUC and TPR@5%FPR (T@F) of PHD and Short-PHD. For results of other baselines, see Tulchinskii et al. (2024).

	Wikipedia						**WritingPrompts**	
	GPT-2		OPT		GPT-3.5		GPT-3.5	
Method	AUC	T@F	AUC	T@F	AUC	T@F	AUC	T@F
PHD	0.825	0.388	0.821	0.423	0.847	0.549	0.808	0.088
Short-PHD	**0.860**	**0.445**	**0.854**	**0.446**	**0.891**	**0.555**	**0.845**	**0.329**
Improve(↑)	4.242	14.691	4.019	5.437	5.195	1.093	4.579	273.864

Table 4: Results on the Generated Dataset. The table reports AUC of baselines and our method on texts (50 tokens). The improvement is computed between PHD and Short-PHD.

Method	Wikihow					Eli5				
	OPT	OPT-iml	NeoX	GPT-3.5	GPT-4o	OPT	OPT-iml	NeoX	GPT-3.5	GPT-4o
RoBERTa	0.66	**0.77**	0.66	0.65	0.66	0.75	0.78	0.77	0.80	0.80
Perplexity	0.47	0.44	0.47	0.62	0.63	0.46	0.37	0.46	0.70	0.73
LogRank	0.46	0.43	0.47	0.58	0.59	0.42	0.33	0.42	0.62	0.64
DetectGPT	0.56	0.53	0.56	0.62	0.59	0.22	0.16	0.21	0.27	0.27
DetectLLM	0.44	0.43	0.41	0.45	0.47	0.37	0.30	0.34	0.36	0.36
PHD	0.62	0.60	0.66	0.66	0.63	0.64	0.65	0.62	0.66	0.66
Short-PHD	**0.78**	0.73	**0.77**	**0.80**	**0.79**	**0.84**	**0.82**	**0.82**	**0.87**	**0.86**
Improve(↑)	25.81	21.67	16.67	21.21	25.40	31.25	26.15	32.26	31.82	30.30

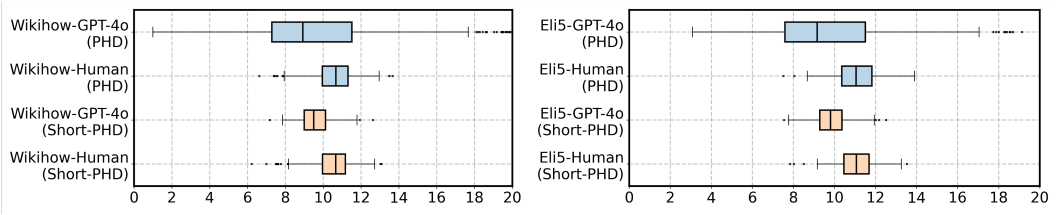


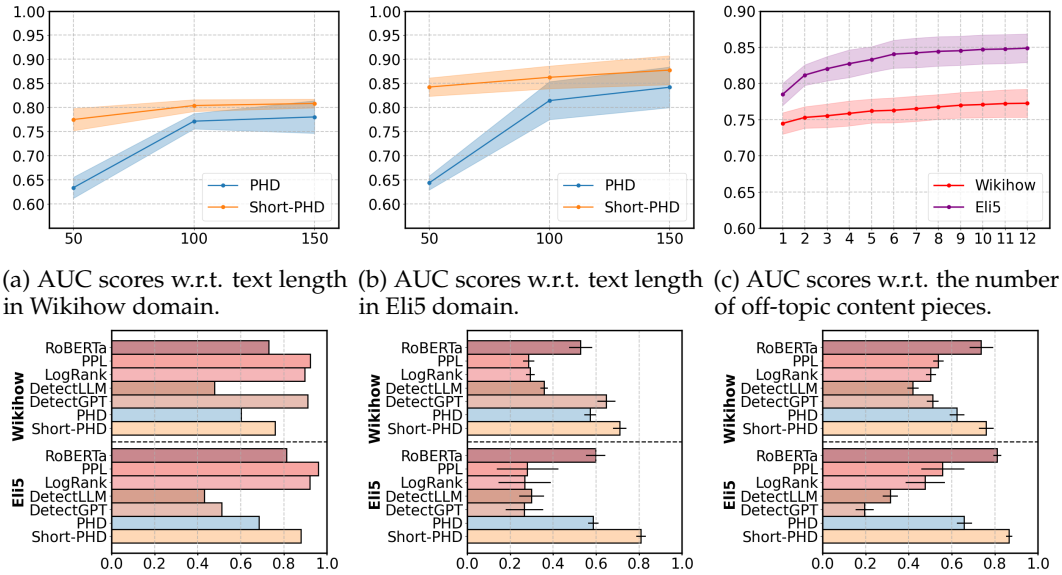
Figure 3: Box plots of PHD and Short-PHD distributions for texts generated by GPT-4o.

We first conduct an empirical evaluation of our method compared to PHD using the public GPTID dataset employed by PHD (without truncation) in terms of AUC and TPR@5%FPR, as shown in Table 3. We observe that Short-PHD outperforms PHD across all four subsets in terms of both metrics. Next, we demonstrate the AUC scores on the generated dataset in Table 4 for Short-PHD and baselines. As reported, our Short-PHD method outperforms other baselines in almost all datasets across five LLMs and two domains, corroborating its effectiveness. Specifically, we observe that detecting short LLM-generated text is generally a challenging task for existing zero-shot methods. For example, DetectGPT fails to detect LLM-generated ELI5 answers, achieving an average AUC of only 0.24. This suggests that probability alone is not a reliable metric for detecting short texts. In contrast, PHD (Tulchinskii et al., 2024) attains an average AUC of approximately 0.65. Similarly, our method achieves an average AUC score of over 0.8, surpassing even the supervised RoBERTa detector. We thus conclude that PHD captures more meaningful information for short text detection through text embeddings. Furthermore, we present box plots of PHD distributions computed by PHD and Short-PHD in Figure 3 for texts generated by GPT-4o. The results for all LLMs are provided in Figure A1 in the appendix. From these figures, we observe that OCI used by Short-PHD increases PHD's value and stabilizes PHD estimations by reducing variance. Meanwhile, the gap between human-written and LLM-generated texts remains largely unchanged, making Short-PHD effective for detection. As a result, Table 4 shows that our method outperforms PHD by more than 16% across all subsets.

4.3 Sensitivity Analysis

We then conduct a series of sensitivity analyses regarding text length, OCI, white-box detection, and detection attacks to further evaluate Short-PHD.

Sensitivity Analysis on Text Length. We first evaluate the detection performance on texts with a fixed length of 50, 100, and 150 tokens in Figures 4a and 4b. As depicted, for texts with a fixed length of 50 tokens, Short-PHD significantly outperforms PHD in terms of AUC. When the text becomes longer, at 100 and 150 tokens, the gap continues to narrow, but Short-PHD still surpasses PHD. The reason is that as the text length increases, the PHD method encounters fewer local density peak problems. To sum up, our proposed Short-PHD



(a) AUC scores w.r.t. text length in Wikihow domain.

(b) AUC scores w.r.t. text length in Eli5 domain.

(c) AUC scores w.r.t. the number of off-topic content pieces.

(d) AUC scores in white-box (LLaMA) detection.

(e) AUC scores under decoherence attack.

(f) AUC scores under paraphrasing attack.

Figure 4: Sensitivity Analysis.

is highly effective for short texts of around 50 tokens and remains superior to the original PHD method for longer texts.

Sensitivity Analysis on the Number of Off-topic Content Pieces. We adjust the number of content pieces in the off-topic content set in Algorithm 1 and analyze its impact. As depicted in Figure 4c, by simply inserting a piece of off-topic content, the average AUC score reaches around 0.75. The performance of Short-PHD stabilizes when at least 6 content pieces are provided. This demonstrates that by inserting multiple off-topic content pieces, the performance of Short-PHD becomes more stable and effective.

Sensitivity Analysis on White-box Detection. We then evaluate the performance of each method in a white-box setting. Specifically, since we use LLaMA as the scoring model for all cases, we generate WikiHow and ELI5 articles using LLaMA with 50 tokens and then evaluate the detection performance. As shown in Figure 4d, probability-based methods such as PPL and DetectGPT achieve the best performance. This is because the probability calculation is more accurate in a white-box scenario. However, the Short-PHD method still achieves an AUC score above 0.9, outperforming PHD and baselines like RoBERTa and DetectLLM, demonstrating its effectiveness in the white-box setting.

Robustness against Detection Attacks. Lastly, we evaluate the robustness of each method against the decoherence attack (Bao et al., 2024) and the paraphrasing attack (Krishna et al., 2023) (see Appendix B.2 for more details) in Figure 4e and Figure 4f. We can observe that Short-PHD remains robust against these attacks, aligning with previous findings that PHD is more robust than other baselines (Tulchinskii et al., 2024).

5 Conclusions and Future Work

In this study, we propose Short-PHD, a novel zero-shot LLM-generated text detection method tailored for short texts. Experimental results demonstrate that Short-PHD achieves superior detection performance for short texts compared to PHD and other baselines. Meanwhile, Short-PHD is robust against various detection attacks. Future work can explore high-dimensional topological structures of embeddings for detection or develop supervised methods based on the detection scores from Short-PHD.